<u>Interview Questions Preparation for TCS NQT Prime 2026</u>

Ques.1 What am I doing currently?

Ans. “Currently, I am working as a Project Trainee at CSIR–CBRI in the domain of Machine Learning under the guidance of a Principal Scientist.

In this project, our goal is to classify different regions of India into zones based on environmental parameters such as mean sea level pressure, vapour pressure, and relative humidity.

We are analysing data from the IMD dataset and identifying key parameters that best represent each state. Based on similarity in these parameters, we group states into zones, which can later be used for region-specific planning and analysis.

So far, we have identified important parameters for regions like Jammu & Kashmir and Himachal Pradesh, and we are currently working on extending this analysis to other states.”

Ques.2 How are you doing it?

Ans. We are doing it in a structured way using data analysis and machine learning techniques.

First, we collect and preprocess the IMD dataset by handling missing values and normalizing the data.

Then, we analyse different environmental parameters like MSLP, vapour pressure, and humidity to identify which features are most relevant for each region.

After that, we use clustering techniques to group states with similar characteristics into zones.

Finally, we evaluate the results and refine the parameters to ensure meaningful and accurate zone formation.

My role involves data preprocessing, feature analysis, and assisting in applying clustering techniques.

Ques.3 Tell me about a time when you handled the conflict in your team?

Ans. During my internship at Suvidha Foundation, I was part of a team of nine members working on optimizing transformer-based models. At one point, a conflict arose within the team—some members preferred using BERT, while others supported Pegasus.

To resolve this, I took the initiative to organize a discussion where both groups could present their perspectives, including the implementation approach, advantages, and limitations of each model.

After evaluating both options and encouraging an open discussion, we conducted a team vote and reached a consensus to proceed with the Pegasus model.

- **What did you learn?**
- I learned the importance of listening to all perspectives and making decisions based on logical evaluation rather than personal preference.

Ques.4 Why do you want to join TCS ?

Ans. TCS is one of the largest IT service companies globally with 6,00,000+ employees across 50+ countries. I’m drawn to the **Ignite Training Program** for freshers – It’s structured and prepares joiners for real client work. I also want to work on projects across different domains, which TCS’s diverse portfolio makes possible. That breadth of experience at this stage of my career is something I won’t find easily elsewhere.

About TCS

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What TCS does?

- ➢ TCS help companies build:
    - Build software
    - Manage it systems
    - Use AI and Data
    - Improve business processes

Ques.5 What are your Strengths and Weaknesses?

Ans. Strength - One of my key strengths is my ability to stay focused and make the best use of available resources. For example, during class 11 when I chose Computer Science, I initially struggled with programming and didn't have a laptop. To overcome this, I started learning from books like Sumita Arora's Python book, practiced by writing code on paper, and used school lab time to implement and test my programs. This experience made me more disciplined, resourceful, and confident in learning new technologies. I would also describe myself as focused and optimistic, which helps me stay consistent while solving problems.

Weakness - One area I'm working on is that I sometimes become so focused on my work that I lose track of time. While this helps me stay productive, it can affect my routine. To improve, I've started managing my time better by setting specific schedules and reminders to maintain a healthy balance.

- ➢ **What did you learn from that situation?**
- ➢ It taught me consistency and that limitations can be managed with the right approach.

Ques.6 Tell me about a time you failed and what did you learned?

Ans. During my current internship, I initially made a mistake while working on data preprocessing. My task was to clean and prepare multiple dataset files, and I decided to merge around 50 files—each containing 10,000 to 50,000 rows—into a single dataset before preprocessing.

However, the merged dataset became extremely large, around 3.7 million rows, which made the preprocessing process inefficient and difficult to manage. This approach also led to errors and confusion in handling the data.

I then realized that a better approach would be to preprocess each file individually first and then merge them afterward. This significantly improved efficiency and made the process more manageable.

From this experience, I **learned** the importance of breaking a complex problem into smaller, manageable parts, which leads to more efficient and optimized solutions.

Ques.7 Tell me about yourself.

Ans. I'm Harsh Saxena, a final-year B.Tech. Computer Science student from Central University of Haryana, I built a strong foundation in data structures and core CS subjects during these 4years.

Currently, I’m working as a Machine Learning Intern at CSIR–CBRI, Roorkee, where I’m involved in a research project on climate clustering and zonation of India under the guidance of Principal Scientist Dr. Kishor S. Kulkarni. Previously, I interned at Suvidha Foundation, where I fine-tuned transformer-based models for abstractive text summarization and also got improved ROUGE scores.

I’m interested in applying my skills to real-world problems, and I’m excited about the opportunity to join TCS, given its scale, diverse projects, and global exposure, which align well with my goal of growing as a software engineer.

Ques.8 Why should we hire you?
Ans. There’s a thought by Chanakya that when a sword is drawn, it makes noise and defeats the enemy, while a shield stays silent but protects. I believe a strong team needs both — people who solve problems quickly and those who bring consistency and reliability.

I see myself as someone who brings dedication, commitment, and a strong sense of responsibility to the team. I ensure that tasks are completed properly, and I’m always willing to put in the effort to learn and improve. Along with that, I have a solid foundation in technical skills, which helps me contribute effectively.

Ques.9 Why you consider you are perfect fit for the job?

Ans. I believe I’m a good fit for this role because I have a strong foundation in core computer science subjects along with hands-on experience through internships and projects in machine learning. I have worked on real-world problems like climate clustering and text summarization, which have helped me develop problem-solving and analytical skills.

I’m also a quick learner and adaptable, which I believe is important in a company like TCS that operates at a global scale. Additionally, my interest in learning new skills, including languages, helps me collaborate better in diverse environments.

Ques.10 What is overfitting and how it can be handled?

Ans. Overfitting occurs when a model learns the training data too well, including noise and unnecessary patterns, which causes it to perform poorly on unseen or test data.

To handle overfitting, we can use several techniques:

- Increasing the size of the dataset
- Using regularization methods like L1 or L2
- Applying techniques like cross-validation
- Reducing model complexity
- Using dropout in neural networks

The goal is to ensure that the model generalizes well to new data rather than just memorizing the training data.

Ques.11

25 IMPORTANT
OOPS CONCEPT
INTERVIEW QUESTIONS

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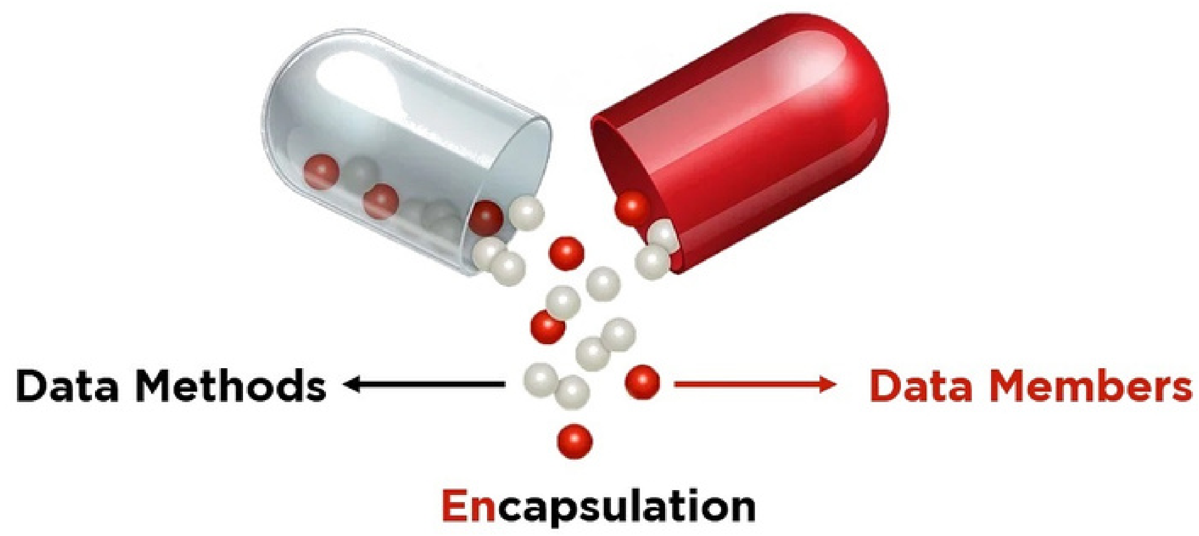
Q 1. What is object-oriented programming (OOPs)?

Ans: Object-Oriented Programming (OOP) is a programming paradigm that organizes code into objects, which are instances of classes.

It focuses on encapsulating data and behavior together, promoting modularity and reusability.

Q 2. What is encapsulation?

Ans: Encapsulation is the concept of bundling data (attributes) and methods (functions) that operate on that data into a single unit (class). It hides the internal details of an object and provides controlled access through methods.



Q 3. What is inheritance?

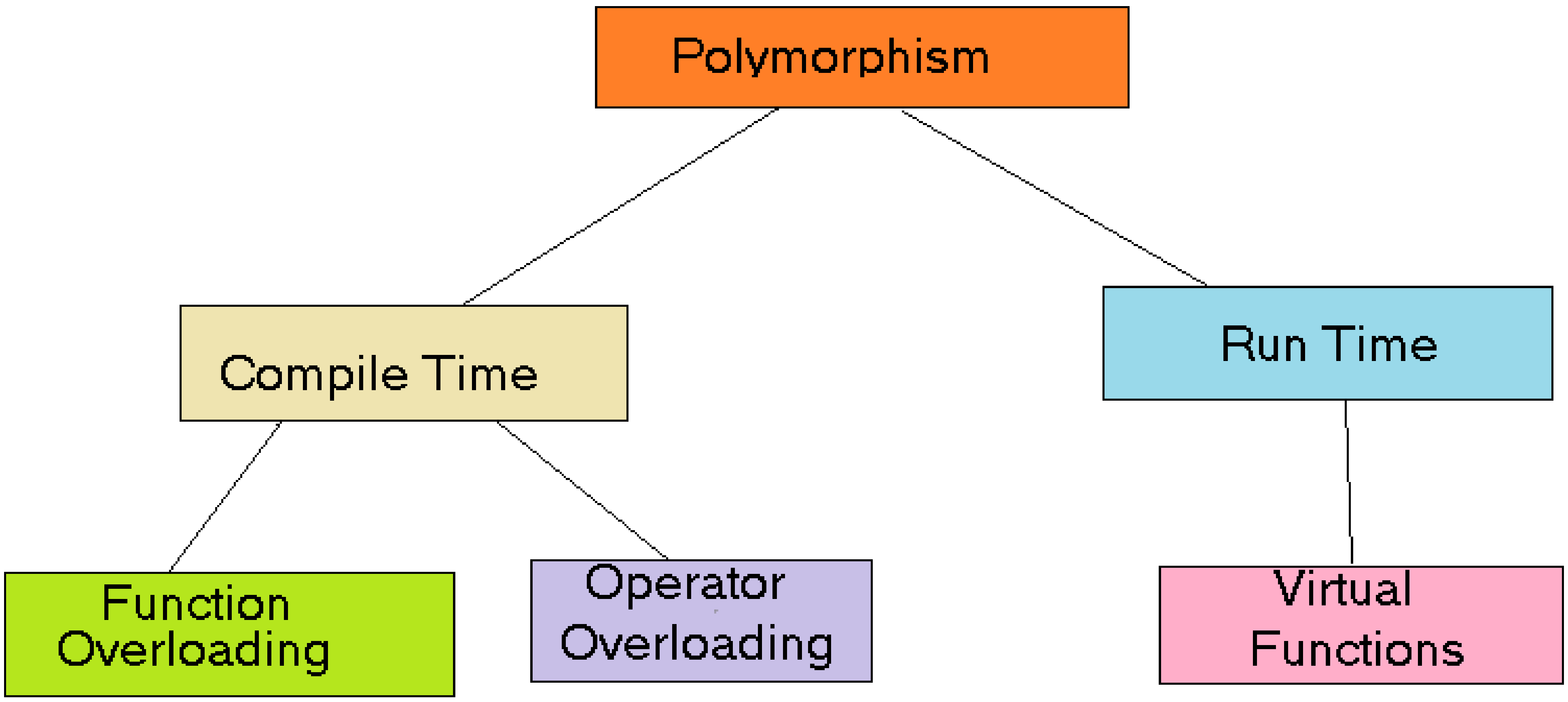
Ans: Inheritance is a mechanism where a new class (subclass or derived class) inherits properties and behaviors from an existing class (superclass or base class).

It promotes code reuse and supports hierarchical relationships.

Q 4. What is polymorphism?

Ans: Polymorphism allows objects of different classes to be treated as instances of a common superclass.

It enables methods to be invoked on objects without knowing their specific types, as long as they adhere to the common interface.



Q 5. What is a constructor?

Ans: A constructor is a special method that is automatically called when an object is created.

It initializes the object's attributes and prepares the object for use. Constructors often have the same name as the class.

Q 6. What is method overloading?

Ans: Method overloading is the ability to define multiple methods in a class with the same name but different parameter lists.

The methods are differentiated based on the number or types of parameters they accept.

Q 7. What is method overriding?

Ans: Method overriding is the process by which a subclass provides a specific implementation for a method that is already defined in its superclass.

The overridden method in the subclass has the same name, return type, and parameters.

Q 8. What is an abstract class?

Ans: An abstract class is a class that cannot be instantiated on its own.

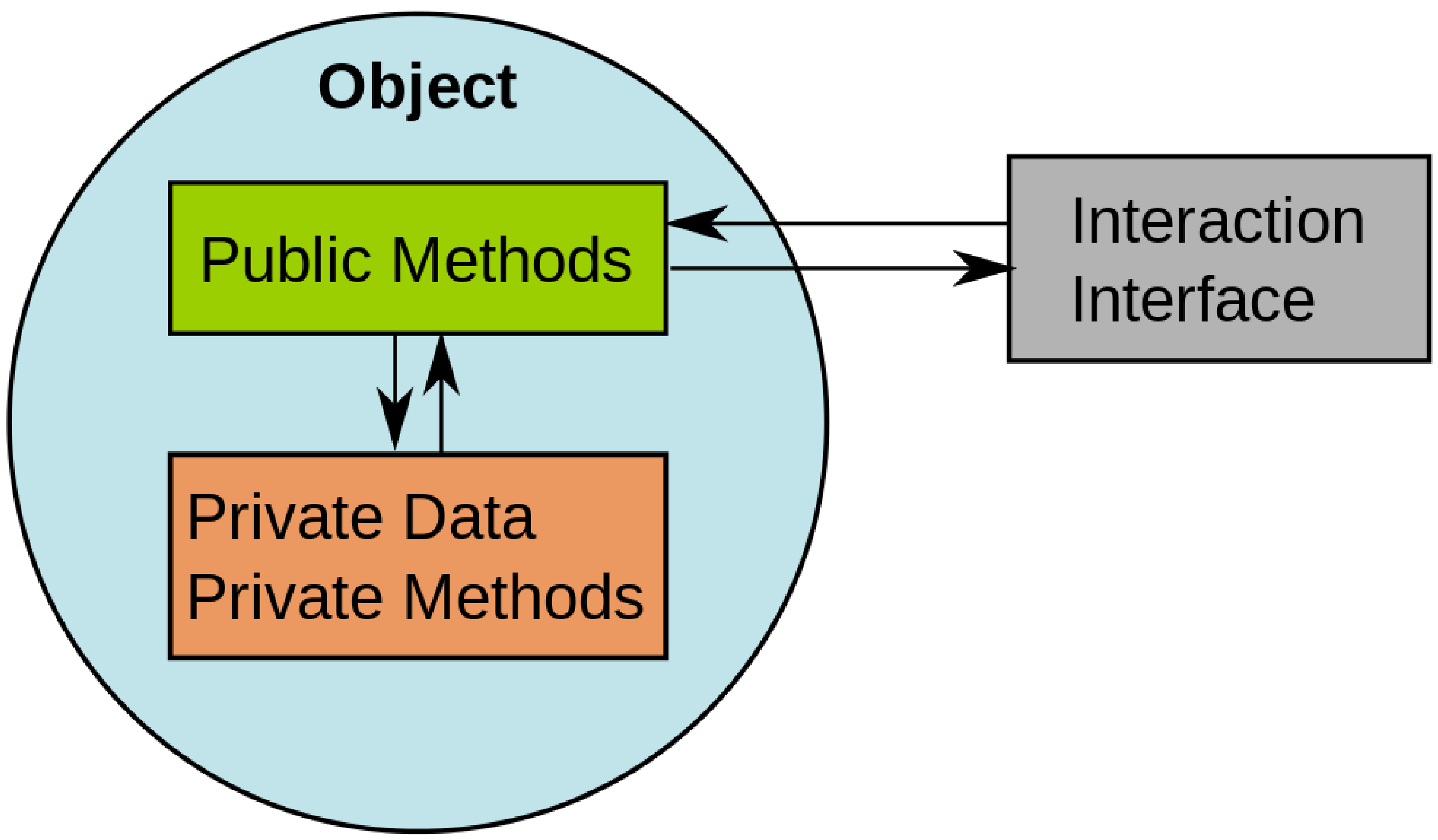
It may contain abstract methods (methods without a body) that must be implemented by its concrete subclasses. Abstract classes provide a common interface.

Q 9. What is an interface?

Ans: An interface is a contract that defines a set of methods that a class must implement.

It allows multiple classes to adhere to the same interface, promoting a form of multiple inheritance.

Interfaces only declare method signatures, not implementations.



Q 10. What is a static method?

Ans: A static method belongs to the class itself, not to instances of the class.

It can be called using the class name and is often used for utility functions or operations that don't require instance-specific data.

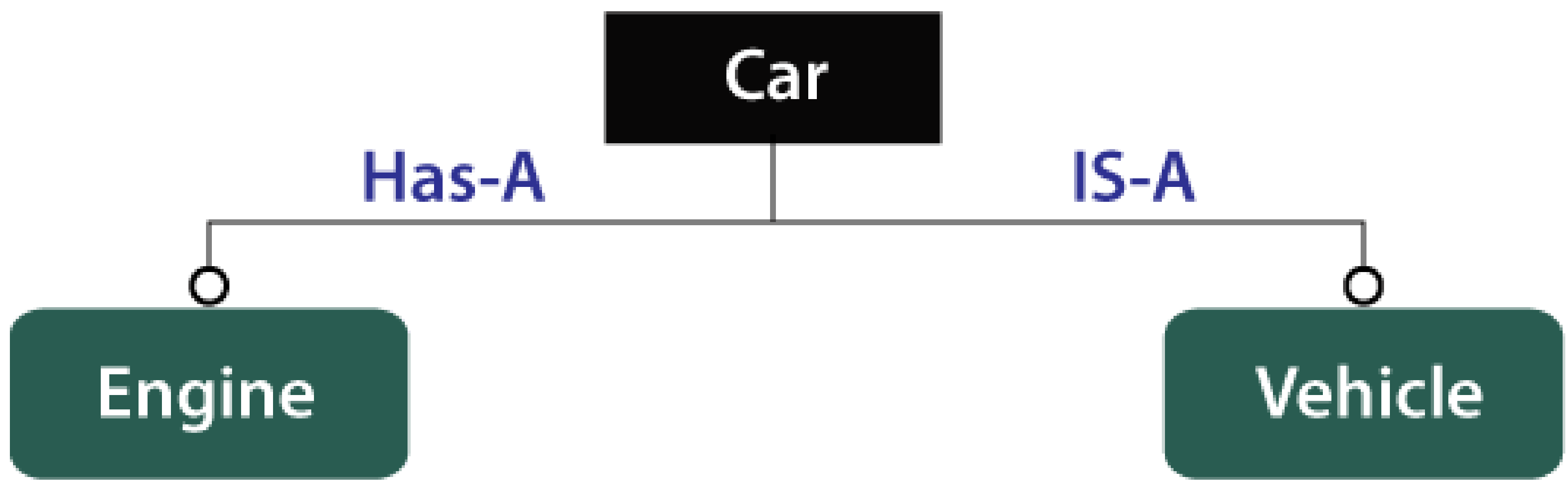
Q 11. What is a final class?

Ans: A final class is a class that cannot be subclassed. It prevents other classes from extending it and inheriting its behavior.

Q 12. What is composition?

Ans: Composition is a design principle where a class contains objects of other classes as part of its attributes.

It allows creating complex structures by combining simpler classes.



Q 13. What is a super keyword?

Ans: The **super keyword** is used to refer to the parent class or superclass. It can be used to call methods and constructors from the superclass.

Q 14. What is method hiding?

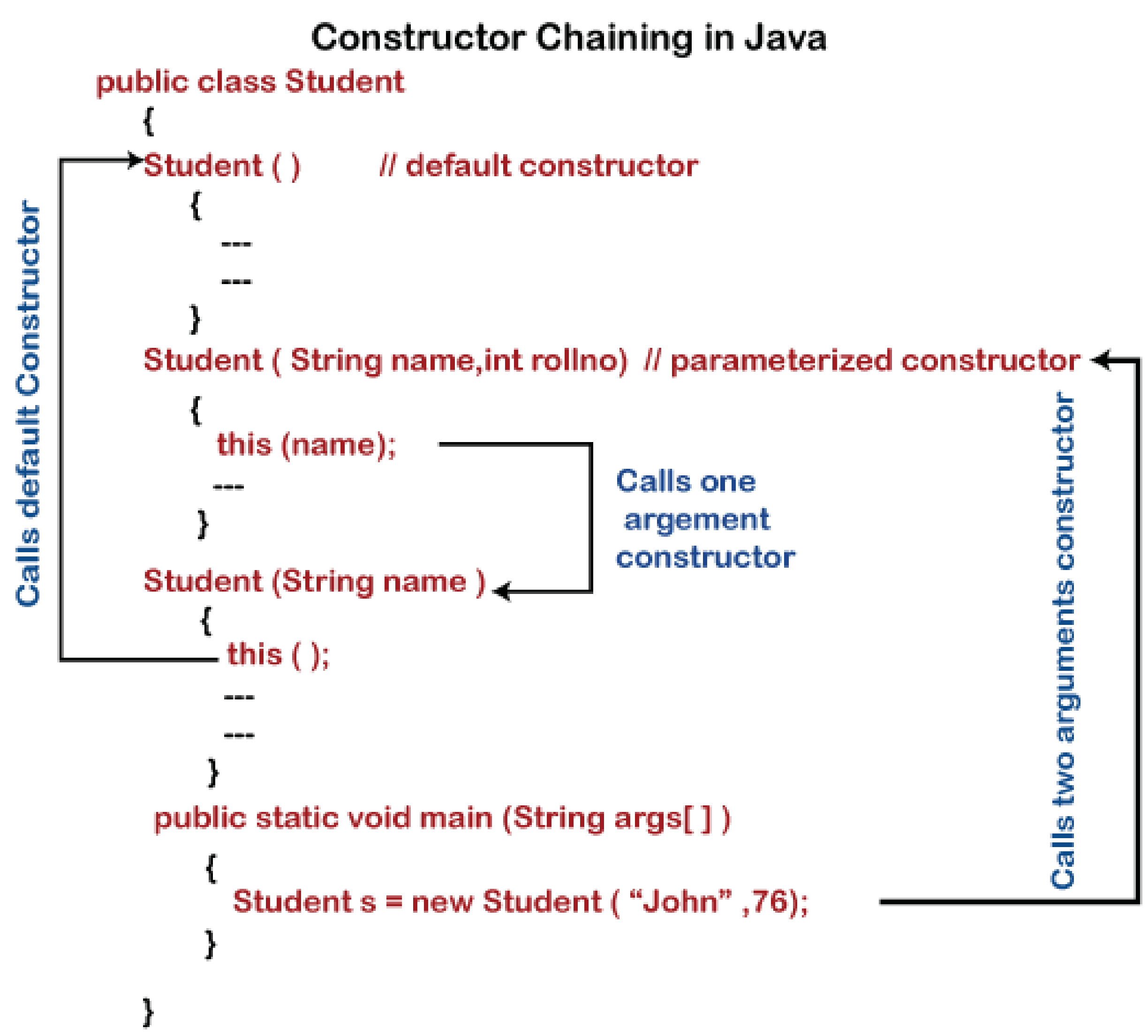
Ans: Method hiding is a concept in object-oriented programming where a subclass defines a method with the same name as a method in its superclass, but the subclass method doesn't override the superclass method.

Instead, it creates a new, independent method with the same name. This can lead to confusion and unexpected behavior, so it's generally recommended to use method overriding instead.

Q 15. What is a constructor chaining?

Ans: Constructor chaining is the process of calling one constructor from another within the same class or between a superclass and a subclass.

It allows for code reuse and efficient initialization.



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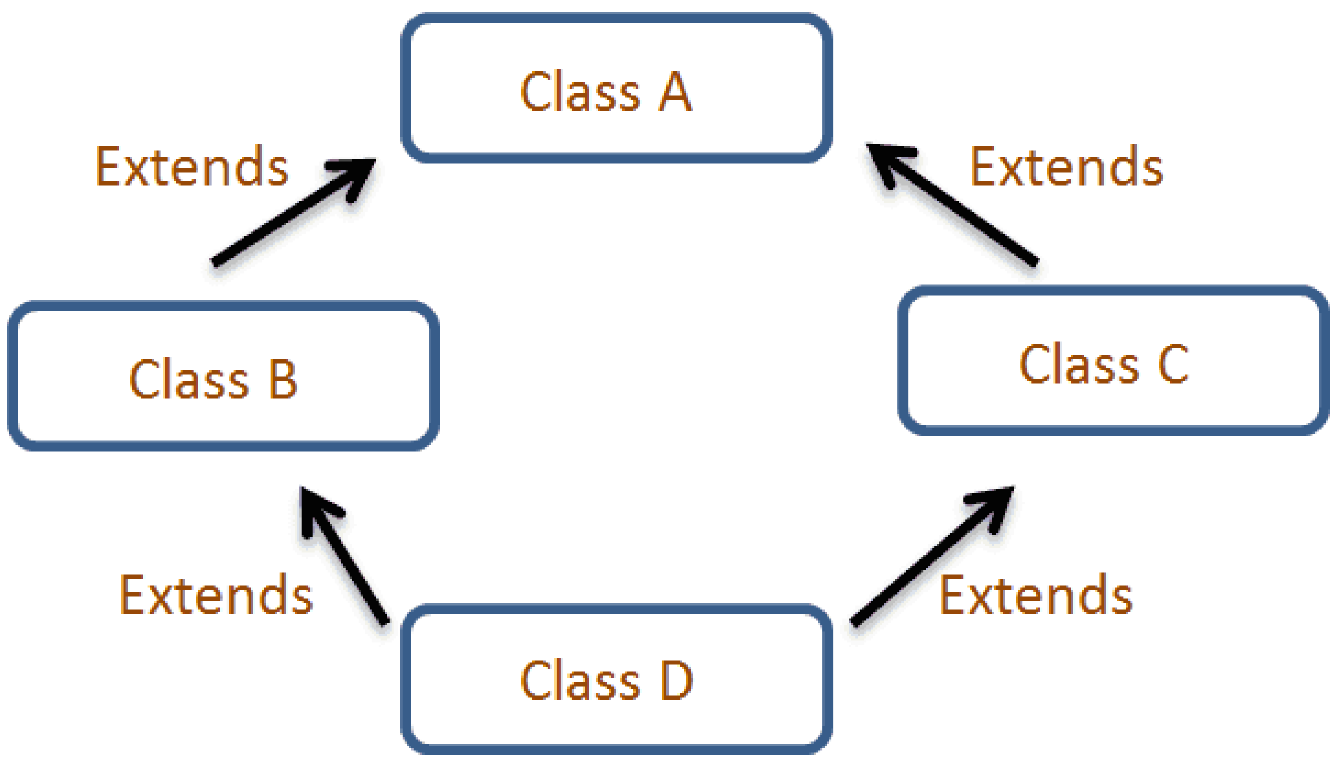
Q 16. What is the diamond problem in multiple inheritance?

Ans: The diamond problem occurs in languages that support multiple inheritance, where a class inherits from two classes that have a common base class.

This can lead to ambiguity in method resolution. Some languages provide mechanisms to handle this, like virtual inheritance.



Q 17. What is a shallow copy and a deep copy?

Ans: A shallow copy copies the references of objects contained within an object, while a deep copy creates new instances of the objects contained.

A deep copy results in a completely independent copy of the original object and its contained objects.

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Q 18. What is a virtual method?

Ans: A virtual method is a method declared in a base class that can be overridden by its subclasses. The actual implementation invoked is determined by the runtime type of the object, supporting polymorphism.

Q 19. How does Java achieve multiple inheritance?

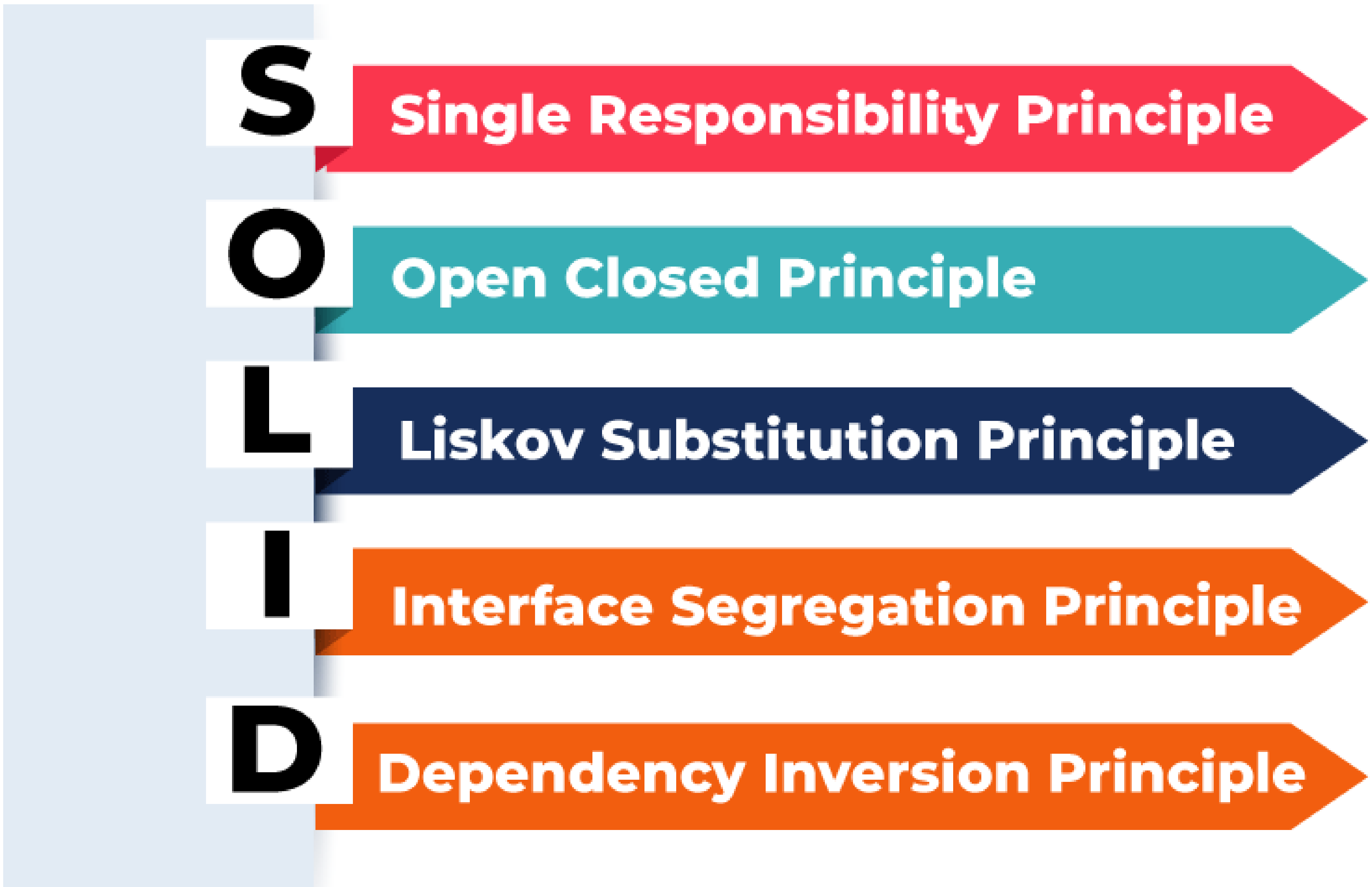
Ans: Java achieves multiple inheritance through interfaces. A class can implement multiple interfaces, allowing it to inherit multiple sets of method declarations.

Q 20. What is the purpose of the final keyword in Java?

Ans: The final keyword can be applied to classes, methods, and variables. A final class cannot be subclassed, **a final** method cannot be overridden, and a final variable cannot be reassigned after its initial assignment.

Q 21. What is the SOLID principle?

Ans: The SOLID principle is an acronym for a set of design principles that promote maintainable and scalable code:



Q 22. What is the difference between an abstract class and an interface?

Ans: An abstract class can have both abstract and concrete methods, while an interface only has method signatures (no method implementations).

A class can implement multiple interfaces, but it can inherit from only one abstract class.

Q 23. How is encapsulation related to data hiding?

Ans: Encapsulation involves bundling data and methods together.

Data hiding is the practice of making the internal data of an object inaccessible to the outside world, except through well-defined methods. Encapsulation helps achieve data hiding.

Q 24. What is the difference between composition and inheritance?

Ans: Inheritance creates a relationship between a subclass and a superclass, allowing the subclass to inherit properties and methods.

Composition involves creating an object within another object to achieve complex behavior without the tight coupling of inheritance.

Q 25. What is the role of a destructor in C++?

Ans: In C++, a destructor is a special method with the same name as the class, preceded by a tilde (~). It is called when an object is about to be destroyed, allowing for cleanup tasks such as releasing resources or memory.

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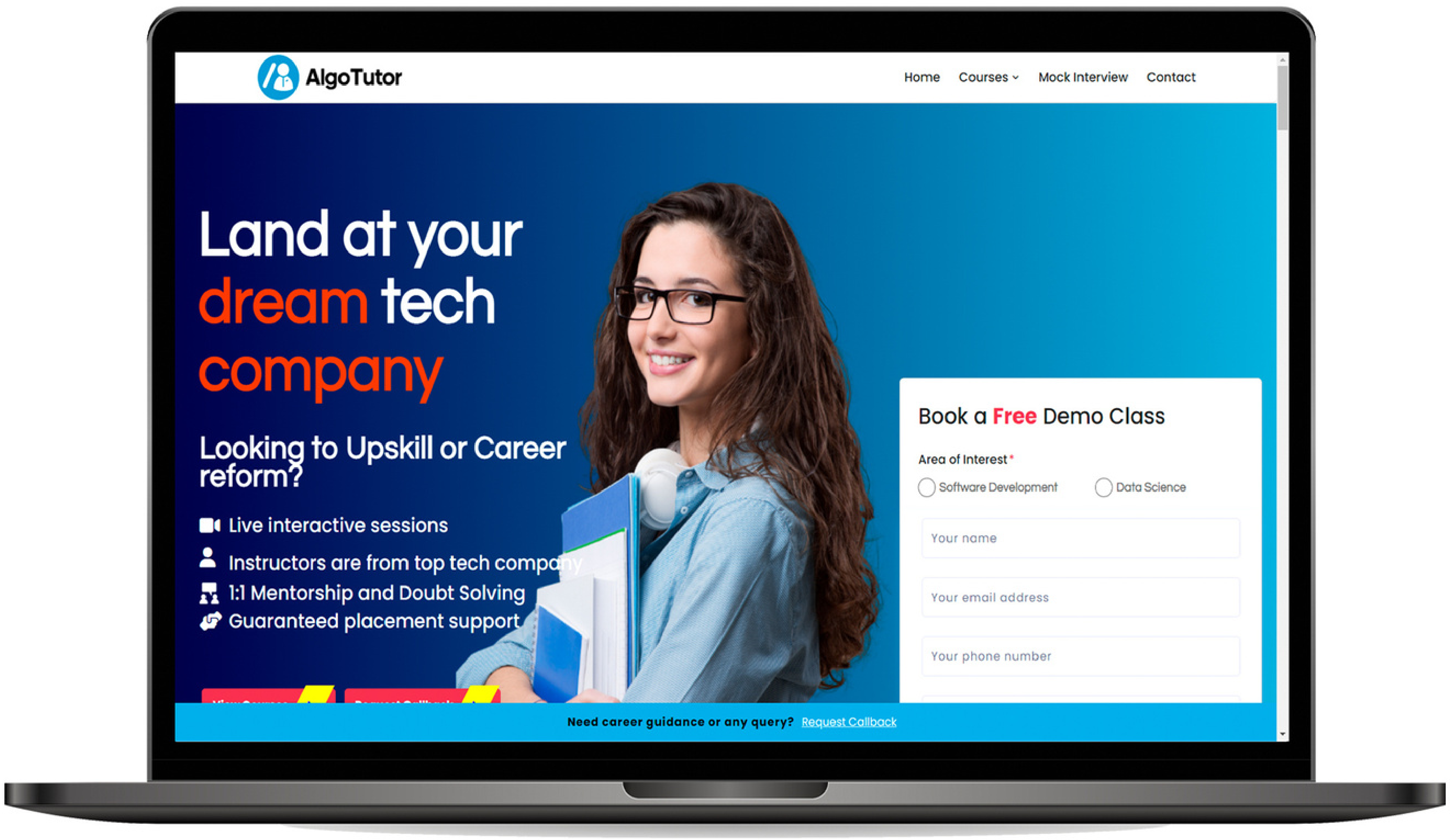

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